

Limits and Continuity

Complex Analysis

Ronny Morales Carmona

Autumn 2025

Complex Functions

Limit Definition

Divergence Criteria

Limit Theorems

Extensions of the
Limit Concept

Continuity

Global Continuity

The connection with
functions of $\mathbb{R}^2$

Continuity
Properties

Definition

A **complex-valued function** is a function with range in $\mathbb{C}$ and a **function of complex variable** is a function whose domain is contained in $\mathbb{C}$. Furthermore, a **complex function** is a complex-valued function of complex variable.

Definition

Let $\Omega \subseteq \mathbb{C}$, and let z_0 be a cluster point of Ω . For a function $f : \Omega \rightarrow \mathbb{C}$, a complex number L is said to be a **limit of f at z_0** if, given any $\varepsilon > 0$, there exists a $\delta > 0$ such that if $z \in \Omega$ and $0 < |z - z_0| < \delta$, then $|f(z) - L| < \varepsilon$.

Remark

If L is a limit of f at z_0 , then we also say that f **converges to L at z_0** . We often write

$$L = \lim_{z \rightarrow z_0} f(z) \quad \text{or} \quad L = \lim_{z \rightarrow z_0} f.$$

We also say that “ $f(z)$ approaches L as z approaches z_0 ”. The symbolism

$$f(z) \rightarrow L \quad \text{as} \quad z \rightarrow z_0$$

is also used sometimes to express the fact that f has a limit L at z_0 .

If the limit of f at z_0 does not exist, we say that f **diverges at z_0** .

Contents

Complex Functions

Limit Definition

Divergence Criteria

Limit Theorems

Extensions of the
Limit Concept

Continuity

Global Continuity

The connection with
functions of $\mathbb{R}^2$

Continuity
Properties

Theorem

Let $f : \Omega \rightarrow \mathbb{C}$ and let z_0 be a cluster point of Ω . Then f can only have one limit at z_0 .

Theorem

Let $f : \Omega \rightarrow \mathbb{C}$ and let z_0 be a cluster point of Ω . Then the following statements are equivalent.

- i) $\lim_{z \rightarrow z_0} f = L$.
- ii) Given any ε -ball $B_\varepsilon(L)$ of L , there exists a δ -ball $B_\delta(z_0)$ of z_0 such that if $z \neq z_0$ is any point in $B_\delta(z_0) \cap \Omega$, then $f(z)$ belongs to $B_\varepsilon(L)$.

Theorem (Sequential Criterion)

Let $f : \Omega \rightarrow \mathbb{C}$ and let z_0 be a cluster point of Ω . Then the following are equivalent.

- i) $\lim_{z \rightarrow z_0} f = L$.
- ii) *For every sequence (z_n) in Ω that converges to z_0 such that $z_n \neq z_0$ for all $n \in \mathbb{N}$, the sequence $(f(z_n))$ converges to L .*

Theorem (Divergence Criteria)

Let $\Omega \subseteq \mathbb{C}$, let $f : \Omega \rightarrow \mathbb{C}$ and let $z \in \mathbb{C}$ be a cluster point of Ω .

- a) If $w \in \mathbb{C}$, then f does **not** have a limit w at z_0 if and only if there exists a sequence (z_n) in Ω with $z_n \neq w$ for all $n \in \mathbb{N}$ such that the sequence (z_n) converges to z_0 but the sequence $(f(z_n))$ does **not** converge to w .
- b) The function f does **not** have a limit at z_0 if and only if there exists a sequence (z_n) in Ω with $z_n \neq w$ for all $n \in \mathbb{N}$ such that the sequence (z_n) converges to z_0 but the sequence $(f(z_n))$ does **not** converge in $\mathbb{R}$.

Definition

Let $\Omega \subseteq \mathbb{C}$, let $f : \Omega \rightarrow \mathbb{C}$, and let $z_0 \in \mathbb{C}$. We say that f is **bounded on a neighborhood of z_0** if there exists a δ -ball $B_\delta(z_0)$ of z_0 and a real number $M > 0$ such that we have $|f(z)| \leq M$ for all $z \in \Omega \cap B_\delta(z_0)$.

Contents

Complex Functions

Limit Definition

Divergence Criteria

Limit Theorems

Extensions of the
Limit Concept

Continuity

Global Continuity

The connection with
functions of $\mathbb{R}^2$

Continuity
Properties

Theorem

If $\Omega \subseteq \mathbb{C}$ and $f : \Omega \rightarrow \mathbb{C}$ has a limit at $z_0 \in \mathbb{C}$, then f is bounded on some neighborhood of z_0 .

Definition

Let $\Omega \subseteq \mathbb{C}$ and let f and g be functions defined on Ω to $\mathbb{C}$. We define the **sum** $f + g$, the **difference** $f - g$, and the **product** fg on Ω to $\mathbb{C}$ to be the functions given by

$$(f + g)(z) := f(z) + g(z), \quad (f - g)(z) := f(z) - g(z),$$

$$(fg)(z) := f(z)g(z),$$

for all $z \in \Omega$. Further, if $\lambda \in \mathbb{C}$, we define the **multiple** λf to be the function given by

$$(\lambda f)(z) := \lambda f(z) \quad \text{for all } z \in \Omega.$$

Finally, if $h(z) \neq 0$ for $z \in \Omega$, we define the **quotient** f/h to be the function given by

$$\left(\frac{f}{h}\right)(z) := \frac{f(z)}{h(z)} \quad \text{for all } z \in \Omega.$$

Theorem

Let $\Omega \subseteq \mathbb{C}$, let f, g be functions on Ω to $\mathbb{C}$, and let $z_0 \in \mathbb{C}$ be a cluster point of Ω . Further, let $\lambda \in \mathbb{C}$.

a) If $\lim_{z \rightarrow z_0} f = L$ and $\lim_{z \rightarrow z_0} g = M$, then:

$$\lim_{z \rightarrow z_0} (f + g) = L + M,$$

$$\lim_{z \rightarrow z_0} (f - g) = L - M,$$

$$\lim_{z \rightarrow z_0} (fg) = LM,$$

$$\lim_{z \rightarrow z_0} \lambda f = \lambda L.$$

b) If $h : \Omega \rightarrow \mathbb{C}$, if $h(z) \neq 0$ for all $z \in \Omega$, and if $\lim_{z \rightarrow z_0} h = H \neq 0$, then

$$\lim_{z \rightarrow z_0} \left(\frac{f}{h} \right) = \frac{L}{H}.$$

Definition

Let $\Omega \subseteq \mathbb{C}$ and let $f : \Omega \rightarrow \mathbb{C}$. If $z_0 \in \mathbb{C}$ is a cluster point of Ω , then we say that f **tends to** ∞ as $z \rightarrow z_0$, and write

$$\lim_{z \rightarrow z_0} f = \infty,$$

if for every $M \in \mathbb{R}$ there is a $\delta > 0$ such that for all $z \in \Omega$ with $0 < |z - z_0| < \delta$, then $|f(z)| > M$.

Definition

Let $\Omega \subseteq \mathbb{C}$, let $f : \Omega \rightarrow \mathbb{C}$ and let $D := \mathbb{C} \setminus \overline{B}_r(0)$. Suppose that $D \subseteq \Omega$ for some $r > 0$. We say that $w \in \mathbb{C}$ is a **limit of f as $z \rightarrow \infty$** , and write

$$\lim_{z \rightarrow \infty} f = w \quad \text{or} \quad \lim_{z \rightarrow \infty} f(z) = w,$$

if given any $\varepsilon > 0$ there exists $R > r$ such that for any $z \in \mathbb{C}$ with $|z| > R$, then $|f(z) - w| < \varepsilon$.

Definition

Let $\Omega \subseteq \mathbb{C}$, let $f : \Omega \rightarrow \mathbb{C}$, and let $z_0 \in \Omega$. We say that f is **continuous at** z_0 if, given any number $\varepsilon > 0$, there exists $\delta > 0$ such that if z is any point of Ω satisfying $|z - z_0| < \delta$, then $|f(z) - f(z_0)| < \varepsilon$.

If f fails to be continuous at z_0 , then we say that f is **discontinuous at** z_0 .

Theorem

A function $f : \Omega \rightarrow \mathbb{C}$ is continuous at a point $z_0 \in \Omega$ if and only if given any ε -ball $B_\varepsilon(f(z_0))$ of $f(z_0)$ there exists a δ -ball $B_\delta(z_0)$ of z_0 such that if z is any point of $\Omega \cap B_\delta(z_0)$, then $f(z)$ belongs to $B_\varepsilon(f(z_0))$, that is,

$$f(\Omega \cap B_\delta(z_0)) \subseteq B_\varepsilon(f(z_0)).$$

- Complex Functions
- Limit Definition
- Divergence Criteria
- Limit Theorems
- Extensions of the Limit Concept
- Continuity
- Global Continuity**
- The connection with functions of $\mathbb{R}^2$
- Continuity Properties

Definition

Let $\Omega \subseteq \mathbb{C}$ and let $f : \Omega \rightarrow \mathbb{C}$. If D is a subset of Ω , we say that f is **continuous on the set D** if f is continuous at every point of D . Further, we say that f is **continuous** if it is continuous at every point of its domain.

- Complex Functions
- Limit Definition
- Divergence Criteria
- Limit Theorems
- Extensions of the Limit Concept
- Continuity
- Global Continuity
- The connection with functions of $\mathbb{R}^2$
- Continuity Properties

Theorem

Let $f : \Omega \rightarrow \mathbb{C}$. Then f is continuous at $x_0 + iy_0 = z_0 \in \mathbb{C}$ if and only if $u(x, y)$ and $v(x, y)$ are both continuous at the point (x_0, y_0) (In the two-dimensional real sense).

Theorem

Let $\Omega \subseteq \mathbb{C}$, let f, g be functions on Ω to $\mathbb{C}$, and let $\lambda \in \mathbb{C}$. Suppose that $z_0 \in \Omega$ and that f and g are continuous at z_0 .

- a) Then $f + g, f - g, fg$ and λf are continuous at z_0 .*
- b) If $h : \Omega \rightarrow \mathbb{C}$ is continuous at $z_0 \in \Omega$ and if $h(z) \neq 0$ for all $z \in \Omega$, then the quotient f/h is continuous at z_0 .*

Corollary

Let $\Omega \subseteq \mathbb{C}$, let f and g be continuous on Ω to $\mathbb{C}$, and let $\lambda \in \mathbb{C}$.

- a) The functions $f + g$, $f - g$, fg and λf are continuous on Ω .
- b) If $h : \Omega \rightarrow \mathbb{C}$ is continuous on Ω and $h(z) \neq 0$ for all $z \in \Omega$, then the quotient f/h is continuous on Ω .

- Complex Functions
- Limit Definition
- Divergence Criteria
- Limit Theorems
- Extensions of the Limit Concept
- Continuity
- Global Continuity
- The connection with functions of $\mathbb{R}^2$
- Continuity Properties

Theorem

Let $\Omega \subseteq \mathbb{C}$, let $f : \Omega \rightarrow \mathbb{C}$, and let $|f|$ be defined by $|f|(z) := |f(z)|$ for $z \in \Omega$.

- a) If f is continuous at a point $z_0 \in \Omega$, then $|f|$ is continuous at z_0 .*
- b) If f is continuous on Ω , then $|f|$ is continuous on Ω .*

- Complex Functions
- Limit Definition
- Divergence Criteria
- Limit Theorems
- Extensions of the Limit Concept
- Continuity
- Global Continuity
- The connection with functions of $\mathbb{R}^2$
- Continuity Properties

Theorem

Let $\Omega, \Psi \subseteq \mathbb{C}$ and let $f : \Omega \rightarrow \mathbb{C}$ and $g : \Psi \rightarrow \mathbb{C}$ be functions such that $f(\Omega) \subseteq \Psi$. If f is continuous at a point $z_0 \in \Omega$ and g is continuous at $w = f(z_0) \in \Psi$, then the composition $g \circ f : \Omega \rightarrow \mathbb{C}$ is continuous at z_0 .

- Complex Functions
- Limit Definition
- Divergence Criteria
- Limit Theorems
- Extensions of the Limit Concept
- Continuity
- Global Continuity
- The connection with functions of $\mathbb{R}^2$
- Continuity Properties

Corollary

Let $\Omega, \Psi \subseteq \mathbb{C}$, let $f : \Omega \rightarrow \mathbb{C}$ be continuous on Ω , and let $g : \Psi \rightarrow \mathbb{C}$ be continuous on Ψ . If $f(\Omega) \subseteq \Psi$, then the composite function $g \circ f : \Omega \rightarrow \mathbb{C}$ is continuous on Ω .